Experiment No-9

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# K-Nearest Neighbours Classifier
# Breast Cancer Classification

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.metrics import classification_report

# Step 1: Load the breast cancer dataset
data = load_breast_cancer()

X = data.data
y = data.target

# Step 2: Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Step 3: Standardize the features
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Step 4: Create the KNN classifier
knn = KNeighborsClassifier(n_neighbors=5)

# Step 5: Train the model
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knn.fit(X_train, y_train)

# Step 6: Predict the test data
y_pred = knn.predict(X_test)

# Step 7: Evaluate the model
print("KNN Classifier - Breast Cancer Dataset")
print("--------------------------------------")

print("Accuracy:", round(accuracy_score(y_test, y_pred) * 100, 2), "%")

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(
   y_test,
   y_pred,
   target_names=data.target_names
))

# Step 8: Predict a sample tumour
sample = [data.data[0]]

sample_scaled = scaler.transform(sample)
prediction = knn.predict(sample_scaled)

print("Sample Tumour Prediction:",
     data.target_names[prediction[0]])

if prediction[0] == 0:
   print("Result: Malignant (Cancerous)")
else:
   print("Result: Benign (Non-cancerous)")
```

Output of the code:-

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KNN Classifier - Breast Cancer Dataset
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Accuracy: 95.61 %

Confusion Matrix:
[[39  3]
 [ 2 70]]

Classification Report:
              precision    recall  f1-score   support

   malignant       0.95      0.93      0.94        42
      benign       0.96      0.97      0.97        72

    accuracy                           0.96       114
   macro avg       0.96      0.95      0.95       114
weighted avg       0.96      0.96      0.96       114

Sample Tumour Prediction: malignant
Result: Malignant (Cancerous)
```